



FACULTY OF ENGINEERING, SCIENCES AND TECHNOLOGY

(Problem Based Learning Lab Assignment)		
CLOs	Blooms Taxonomy	GAs
CLO1 CLO2 CLO3	P3 C4 A3	GA2 GA3 GA6

Assigned Week	7 th Week	Assessment Week	14 th Week
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WP	Characteristic	Description
WP1 WP3 WP4	Range of conflicting requirements Depth of knowledge required Familiarity of issues	Balancing data collection, data cleaning, mining techniques, and visualization requirements in a data warehousing and mining system. Requires expertise in data warehousing concepts, ETL processes, data mining algorithms, and database management tools. Involves handling large datasets, missing or inconsistent data, mining accuracy, and performance optimization issues.

Name	Abdullah Shah Alam.
ID	IU04-123-0462.
Course	Data Warehousing & Data Mining (Lab).
Instructor	Hafiza Amna Owais Ansari.
Submission Date	11 June 2026.

Consumer Load Pattern Analysis and Demand Forecasting for Power Distribution Companies (DISCOs)

1. Problem Statement

Electric Distribution Companies (DISCOs) like K-Electric are responsible for supplying electricity to millions of consumers across different geographic zones. Managing electricity demand efficiently is a critical challenge for these companies. Without proper analysis of historical consumption data, DISCOs cannot predict peak demand periods, identify high-consumption zones, or make informed load management decisions — often resulting in unplanned load shedding and inefficient resource planning.

Consumer electricity usage varies significantly based on area type, time of day, season, and weather conditions. These patterns are difficult to detect manually from raw meter data. A data-driven approach using Data Warehousing and Data Mining techniques is needed to extract meaningful insights from large-scale smart meter datasets.

A centralized Data Warehousing and Mining system is necessary to:

- Store and manage large-scale smart meter consumption data efficiently.
- Organize structured data across consumer zones, time periods, and weather conditions.
- Perform analytical queries to identify demand trends and consumption patterns.
- Apply data mining algorithms to discover hidden patterns and forecast demand.
- Support better load management and supply planning decisions for DISCOs.

2. Objective

The objective of this project is to design and implement a Consumer Load Pattern Analysis System using Data Warehousing and Data Mining techniques with MS SQL Server 2022, Weka, and Power BI.

The system will:

- Store household-level electricity consumption data in a structured SQL database.
- Manage and relate consumption, household, and weather data using proper table design.
- Handle data quality issues — including null values and inconsistent records — through SQL-based ETL operations.
- Perform 15 analytical SQL queries to extract consumption trends, peak hours, and area rankings.
- Apply K-Means Clustering to group consumer zones by usage behavior.
- Apply J48 Decision Tree Classification to predict high or low demand periods.
- Apply Apriori Association Rules to discover relationships between time, weather, and demand.
- Visualize findings through an interactive Power BI dashboard with 6 visuals.

The project also aims to improve practical understanding of:

- Data warehouse architecture and star schema design
- ETL (Extract, Transform, Load) processes for large datasets
- SQL Server bulk data import and null handling strategies
- Data mining techniques using Weka (clustering, classification, association)
- Business intelligence dashboards using Power BI

3. Features and Functionalities

Main Features

1. Electricity Consumption Data Management

- Store half-hourly smart meter readings for thousands of households.
- Maintain structured records in the consumption, households, and weather tables.
- Use staging table approach to handle Null values during bulk CSV import.

2. Area-wise Consumer Zone Classification

- Store household socioeconomic area group (Acorn Group) as a proxy for K-Electric distribution zones.
- Support zone-level aggregation and comparison queries.
- Enable tariff type comparison (Standard vs Time-of-Use consumers).

3. Data Analysis and Reporting (SQL Queries)

Generate reports for:

- Total and average consumption per area group
- Peak hour and peak day identification
- Monthly and seasonal consumption trends
- Top 10 highest consuming households
- Weekday vs weekend consumption comparison
- Anomaly detection for abnormally high consumption days
- Weather vs consumption correlation

4. ETL Operations

- Extract raw CSV data from Kaggle Smart Meters in London dataset.
- Transform data in Excel — structural cleaning only (column deletion and renaming).
- Load data into SQL Server using BULK INSERT with staging table for null filtering.
- Ensure organized and consistent storage across all three tables.

5. Data Mining Operations (Weka)

- K-Means Clustering: Group households into Low / Medium / High consumption clusters.
- J48 Decision Tree: Classify time periods as high or low demand using area and weather features.
- Apriori Association Rules: Discover patterns such as Evening + Winter = High Load.

6. Power BI Dashboard

- Monthly Consumption Trend — Line Chart
- Area-wise Total Consumption — Bar Chart
- Consumer Segment Distribution — Donut Chart
- KPI Cards — Total kWh, Total Households, Average Consumption
- Weather vs Consumption — Scatter Plot (optional)

4. Design and Implementation

Database Design

The database disco_analysis was created in MS SQL Server 2022. It contains three main tables:

Table Name	Columns	Purpose
consumption	household_id, reading_datetime, energy_kwh	Main fact table — half-hourly electricity readings per household
households	household_id, tariff_type, acorn_code, area_group	Dimension table — area group and tariff classification per household
weather	weather_date, temp_max, temp_min, humidity, wind_speed, cloud_cover, weather_summary, weather_icon	Dimension table — daily weather data for demand correlation analysis

Data Source

- Dataset: Smart Meters in London — Kaggle (Jean-Michel D.)
- Reason for use: Real K-Electric data is not publicly available. This dataset provides equivalent smart meter structure with area-based consumer classification.
- Acorn Groups in dataset = K-Electric Distribution Zones (project framing)
- Data period: November 2011 to February 2014
- Main file used: block_0.csv (57.48 MB, half-hourly readings)

Tools and Technologies

Tool	Purpose
MS SQL Server 2022 + SSMS 20	Database creation, table design, BULK INSERT, SQL analytical queries
Microsoft Excel	Structural data cleaning — column deletion and renaming only
Weka (Latest)	K-Means Clustering, J48 Decision Tree, Apriori Association Rules
Power BI Desktop	Interactive dashboard with 6 visual analytics
Kaggle	Open-source dataset: Smart Meters in London

5. Work Flow & ER Diagram

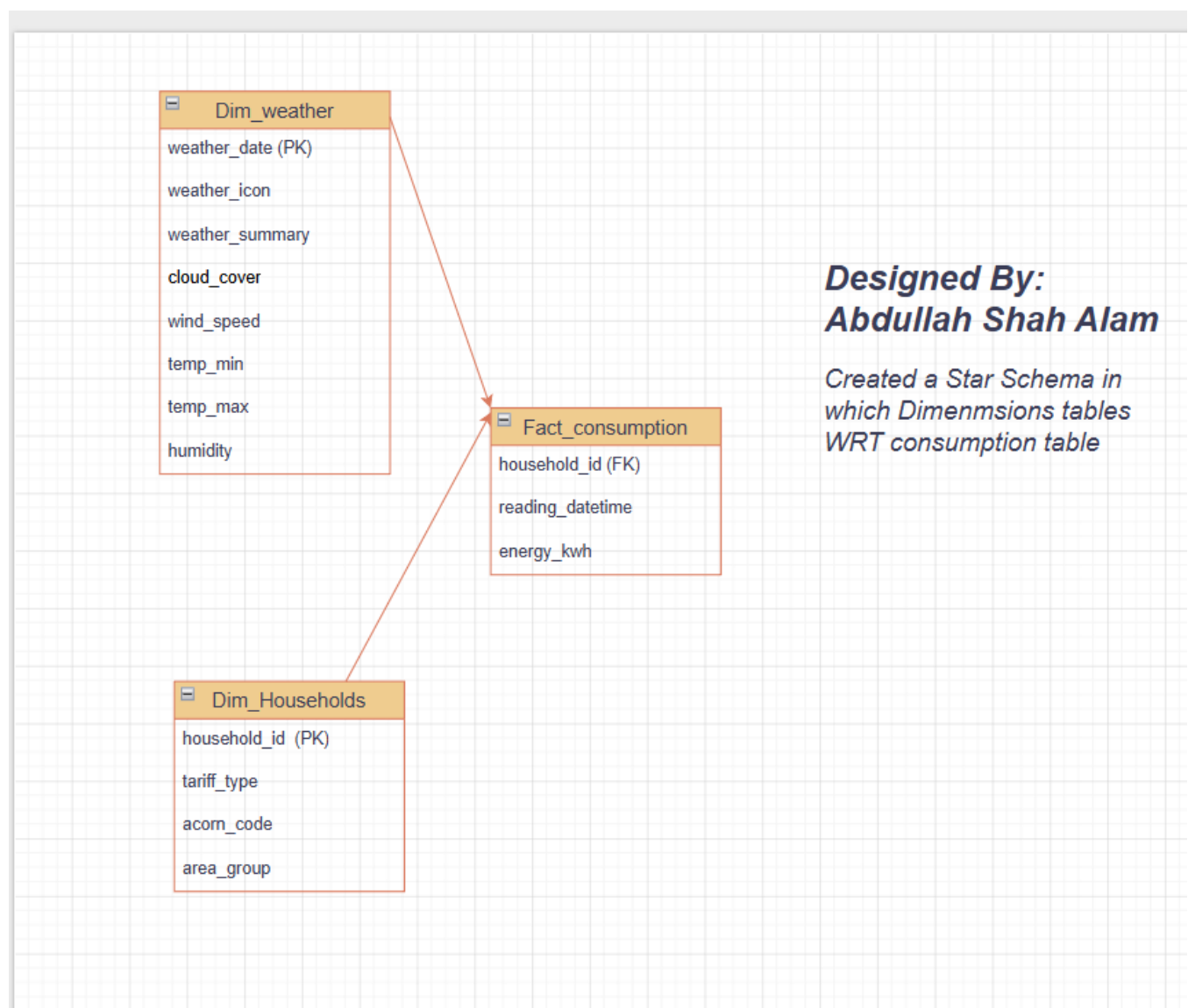
Work Flow

- Download Smart Meters in London dataset from Kaggle (block_0.csv, informations_households.csv, weather_daily_darksky.csv).
- Clean data in Excel — delete unnecessary columns, rename remaining columns to standard names. No null removal in Excel (large file crash risk).
- Create database disco_analysis in MS SQL Server 2022 using SSMS 20.
- Create three tables: consumption, households, and weather with appropriate data types.
- Import households.csv and weather.csv directly via BULK INSERT.
- Import consumption.csv using a staging table approach — raw data loaded into consumption_staging, then filtered (nulls removed, invalid dates excluded) before inserting into the final consumption table.

- Execute 15 SQL analytical queries covering consumption trends, peak hours, area rankings, weather correlation, and anomaly detection.
- Export area-level summary data from SQL as CSV for Weka input.
- Apply K-Means Clustering in Weka to group consumer zones into Low / Medium / High usage clusters.
- Apply J48 Decision Tree in Weka to classify demand periods as high or low.
- Apply Apriori Association Rules in Weka to find patterns between time, weather, and consumption.
- Connect Power BI Desktop to SQL Server database directly.
- Build 6-visual interactive dashboard: Line chart, Bar chart, Donut chart, KPI Cards, and optional Scatter plot.
- Document all results — SQL screenshots, Weka outputs, and Power BI dashboard screenshots — in the final report.

Project Architecture Diagram

The database follows a Star Schema design where the consumption table acts as the central fact table, and households and weather serve as dimension tables.



RESULTS AND OUTPUT

1. SQL

Import 5566 Records from households.csv file into database

```
-- Table 2: Households (area info)
CREATE TABLE households (
    household_id VARCHAR(20),
    tariff_type VARCHAR(10),
    acorn_code VARCHAR(20),
    area_group VARCHAR(20)
);
BULK INSERT households
FROM 'C:\Users\abdul\Desktop\DataWare Project\Data\Cleaned\households.csv'
WITH (
    FIRSTROW = 2,
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '\n',
    TABLOCK
);
```

(5566 rows affected)

Completion time: 2026-06-13T02:50:48.1532270+05:00

Query executed successfully.

select * from households

	household_id	tariff_type	acorn_code	area_group
1	MAC005492	ToU	ACORN-	ACORN-
2	MAC001074	ToU	ACORN-	ACORN-
3	MAC000002	Std	ACORN-A	Affluent
4	MAC003613	Std	ACORN-A	Affluent
5	MAC003597	Std	ACORN-A	Affluent
6	MAC003579	Std	ACORN-A	Affluent
7	MAC003566	Std	ACORN-A	Affluent
8	MAC003557	Std	ACORN-A	Affluent
9	MAC003553	Std	ACORN-A	Affluent
10	MAC003482	Std	ACORN-A	Affluent
11	MAC003463	Std	ACORN-A	Affluent
12	MAC003449	Std	ACORN-A	Affluent
13	MAC003428	Std	ACORN-A	Affluent

ABDULLAH-PC\abdul (58) | disco_analysis2 | 00:00:00 | 5,566 rows

Import 882 Records from wheather.CSV file into database

```
-- Table 3: Weather (daily weather)
CREATE TABLE weather (
    temp_max          FLOAT,
    weather_icon       VARCHAR(50),
    cloud_cover        FLOAT,
    wind_speed         FLOAT,
    humidity           FLOAT,
    weather_date        DATETIME2,
    temp_min           FLOAT,
    weather_summary     VARCHAR(100)
);

BULK INSERT weather
FROM 'C:\Users\abdul\Desktop\DataWare Project\Data\Cleaned\weather.csv'
WITH (
    FIRSTROW = 2,
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '\n',
    TABLOCK
);
```

151 %

Messages

(882 rows affected)

Completion time: 2026-06-13T03:04:17.6874230+05:00

151 %

Query executed successfully. ABDULLAH-PC\SQLABDU

```
select * from weather
```

151 %

Results Messages

	temp_max	weather_icon	cloud_cover	wind_speed	humidity	weather_date	temp_min	weather_summary
1	11.96	fog	0.79	3.88	0.95	2011-11-11 00:00:00.0000000	8.85	Foggy until afternoon.
2	8.59	partly-cloudy-day	0.56	3.94	0.88	2011-12-11 00:00:00.0000000	2.48	Partly cloudy throughout the day.
3	10.33	partly-cloudy-day	0.85	3.54	0.74	2011-12-27 00:00:00.0000000	8.03	Mostly cloudy throughout the day.
4	8.07	wind	0.32	3	0.87	2011-12-02 00:00:00.0000000	2.56	Partly cloudy throughout the day and breezy over...
5	8.22	partly-cloudy-night	0.37	4.46	0.8	2011-12-24 00:00:00.0000000	3.17	Mostly cloudy throughout the day.
6	7.97	wind	0.42	4.71	0.77	2011-12-15 00:00:00.0000000	4.08	Partly cloudy throughout the day and breezy in th...
7	13.19	fog	0.26	2.37	0.92	2011-11-19 00:00:00.0000000	7.48	Foggy starting in the evening.
8	8.32	fog	0.81	2.36	0.93	2011-11-16 00:00:00.0000000	5.19	Foggy starting in the evening.
9	9.82	wind	0.38	5.02	0.84	2011-12-12 00:00:00.0000000	3.09	Partly cloudy throughout the day and breezy start...
10	9.71	fog	0.41	1.24	0.96	2011-11-20 00:00:00.0000000	4.76	Foggy in the morning.
11	11.31	fog	0.83	2.4	0.92	2011-11-08 00:00:00.0000000	9.37	Foggy in the morning.
12	9.11	partly-cloudy-day	0.36	4.31	0.82	2011-12-04 00:00:00.0000000	4.69	Partly cloudy until evening.
13	4.68	partly-cloudy-day	0.7	3.71	0.88	2011-12-16 00:00:00.0000000	1.8	Mostly cloudy throughout the day.
14	15.57	partly-cloudy-day	0.36	2.45	0.84	2011-11-01 00:00:00.0000000	9.68	Partly cloudy until evening.
15	12.91	partly-cloudy-day	0.81	5.44	0.84	2011-12-26 00:00:00.0000000	9.82	Mostly cloudy throughout the day.
16	12.58	partly-cloudy-night	0.56	2.34	0.86	2011-11-28 00:00:00.0000000	1.31	Mostly cloudy throughout the day.
17	7.68	partly-cloudy-day	0.15	5.65	0.71	2011-12-09 00:00:00.0000000	2.03	*Partly cloudy starting in the afternoon, continuin...
18	6.87	partly-cloudy-day	0.2	5.57	0.77	2011-12-05 00:00:00.0000000	2.81	Partly cloudy until evening.
19	12.93	partly-cloudy-night	0.41	4.04	0.89	2011-11-24 00:00:00.0000000	8.56	Partly cloudy throughout the day.
20	14.44	fog	0.52	2.23	0.88	2011-11-17 00:00:00.0000000	8.54	Foggy in the morning.
21	12.08	wind	0.42	7.44	0.75	2011-12-13 00:00:00.0000000	4.54	Breezy in the morning and partly cloudy through...
22	12.14	partly-cloudy-night	0.38	4.31	0.87	2011-12-22 00:00:00.0000000	8.08	Partly cloudy throughout the day.
23	6.91	partly-cloudy-day	0.34	4.79	0.83	2011-12-06 00:00:00.0000000	2.68	Partly cloudy until evening.
24	9.02	wind	0.29	7.06	0.68	2011-12-07 00:00:00.0000000	4.91	Partly cloudy throughout the day and breezy until...
25	10.69	fog	0.7	2.62	0.95	2011-11-21 00:00:00.0000000	7.1	Foggy starting in the evening.
26	12.89	wind	0.53	7.01	0.81	2011-12-08 00:00:00.0000000	4.27	Breezy and mostly cloudy throughout the day.
27	11.85	partly-cloudy-day	0.77	4.77	0.81	2011-11-06 00:00:00.0000000	9.46	Mostly cloudy throughout the day.

Query executed successfully. ABDULLAH-PC\SQLABDULLAHDEVSV... ABDULLAH-PC\abdul (58) d

Ln 92 Col 1 Ch 1 INS

Import 1414074 Records from consumption.CSV file into Staqqing table

```
-- Table 1: Consumption (main data)
CREATE TABLE consumption (
    household_id    VARCHAR(20),
    reading_datetime DATETIME2,
    energy_kwh       FLOAT
);

-- Step 1: Create a Staging table for filter NULL values
CREATE TABLE consumption_staging (
    household_id    VARCHAR(20),
    reading_datetime VARCHAR(50),
    energy_kwh       VARCHAR(20)
);

-- Step 2: Raw data import into staging table
BULK INSERT consumption_staging
FROM 'C:\Users\abdul\Desktop\DataWare Project\Data\Cleaned\block_5.csv'
WITH (
    FIRSTROW = 2,
    FIELDTERMINATOR = ','
);
```

151 %

Messages

(1414074 rows affected)

Completion time: 2026-06-13T04:11:51.4870018+05:00

151 %

Query executed successfully. | ABDULLAH-PC\SQLABDU

After Cleaned Null Values from staging table, shift cleaned data into main table 'consumption' so it is Clearly shows the difference of 50 records were cleaned.

```
-- Step 3: Clean data, Remove Duplicates, Standardize, and Load into Final Table
INSERT INTO consumption (household_id, reading_datetime, energy_kwh)
SELECT DISTINCT
    UPPER(household_id), -- Standardize
    TRY_CAST(reading_datetime AS DATETIME2), -- TRY_CAST escape from error during insertion
    TRY_CAST(energy_kwh AS FLOAT)
FROM consumption_staging
WHERE energy_kwh != 'Null'
AND energy_kwh IS NOT NULL
AND energy_kwh != ''
AND TRY_CAST(reading_datetime AS DATETIME2) IS NOT NULL; -- Sirf valid dates andar jayengi
```

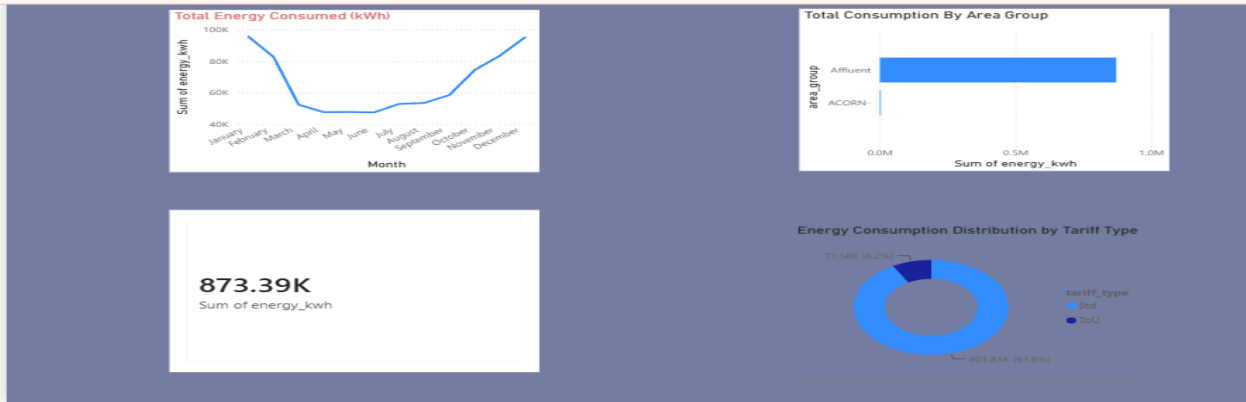
151 %

Messages

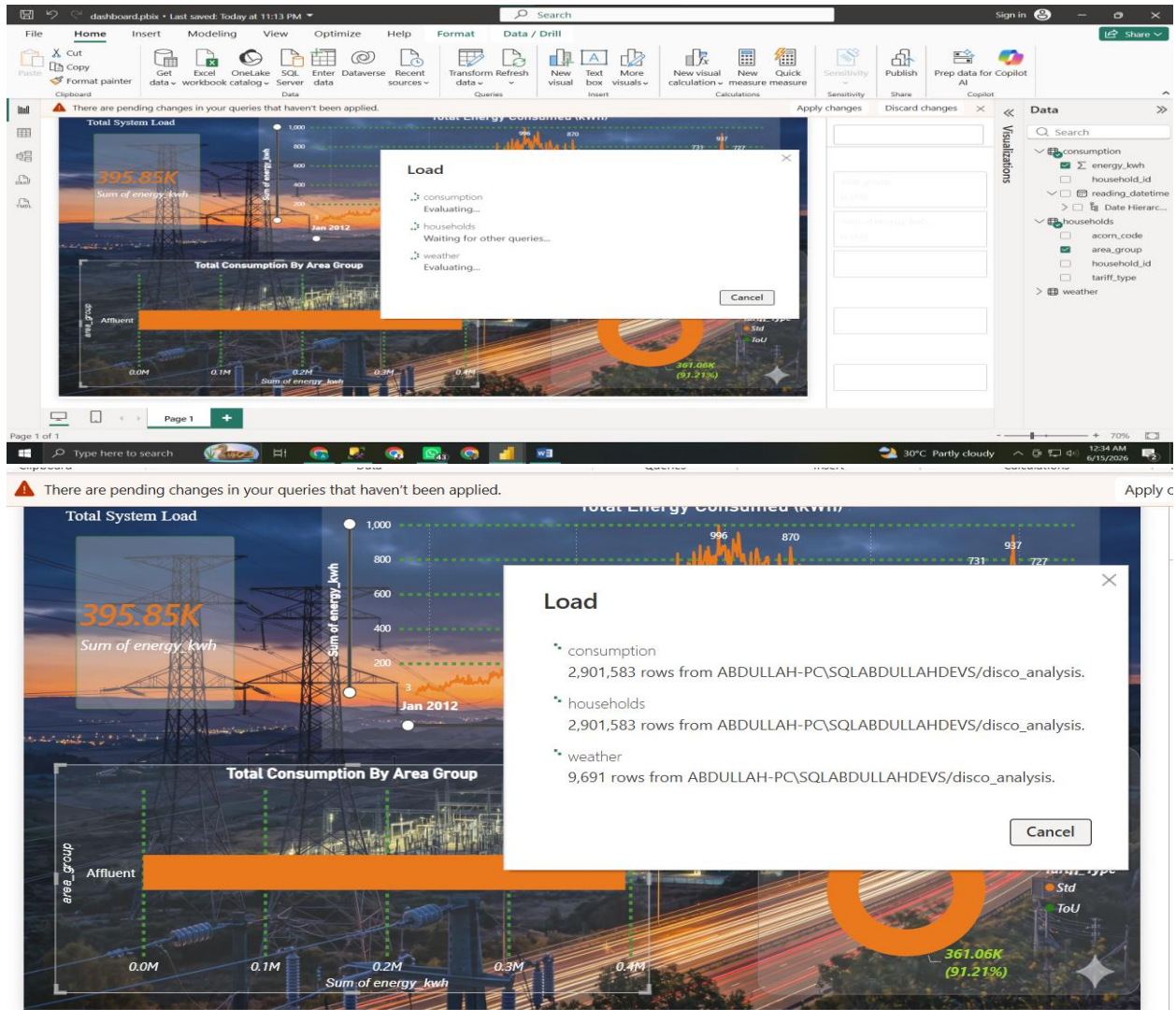
(1414024 rows affected)

Completion time: 2026-06-13T04:13:49.3347322+05:00

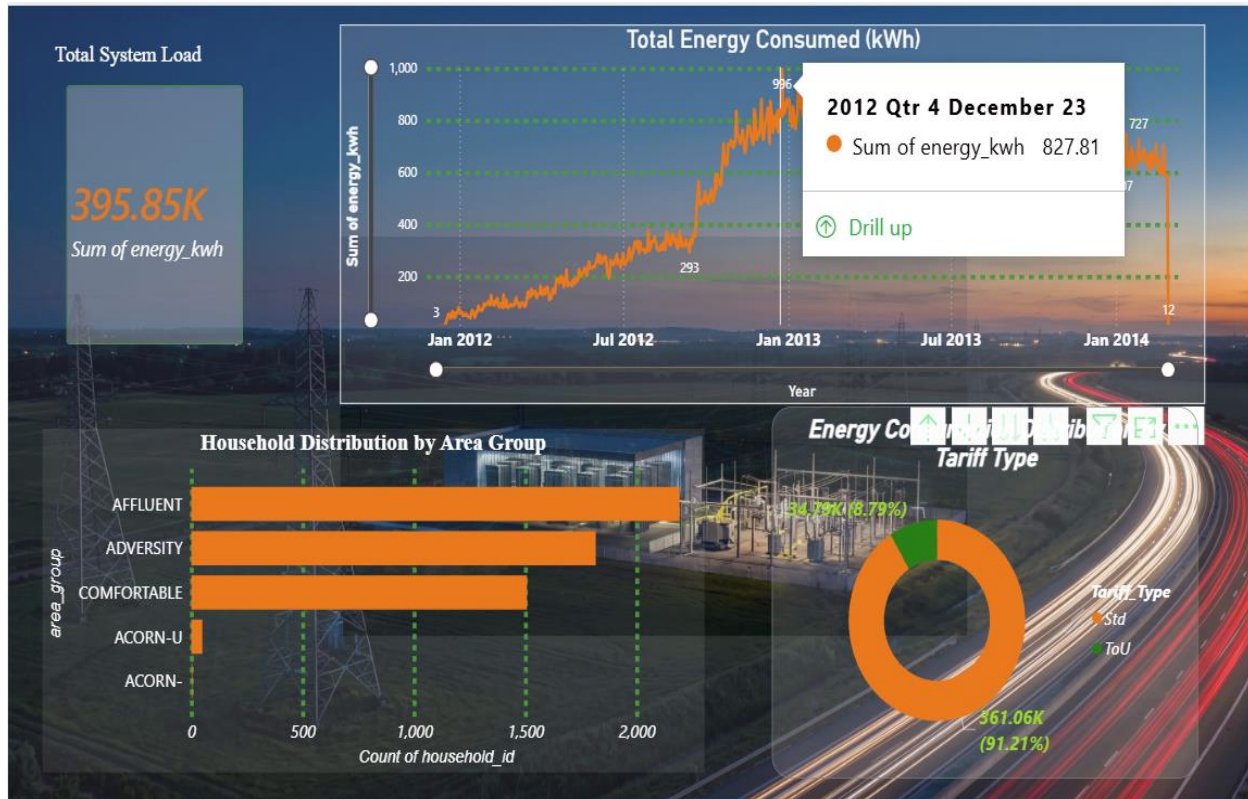
2. POWER BI



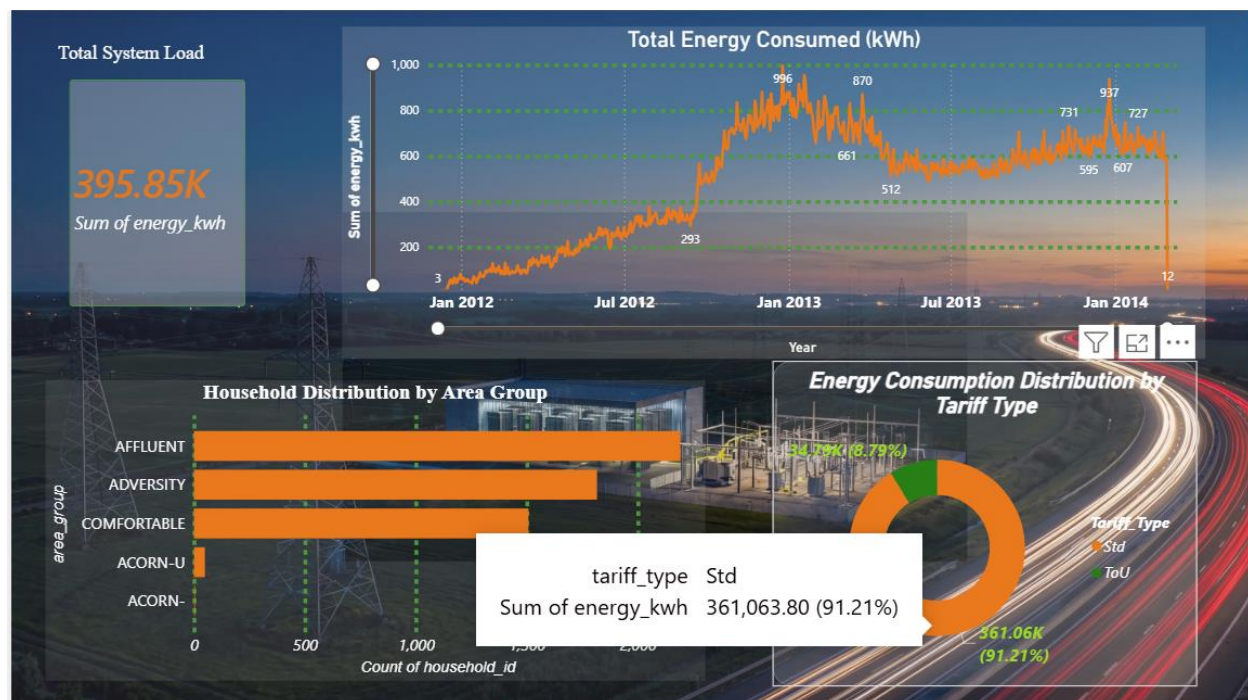
EXECUTION OF TRANSFORMS OF DATA



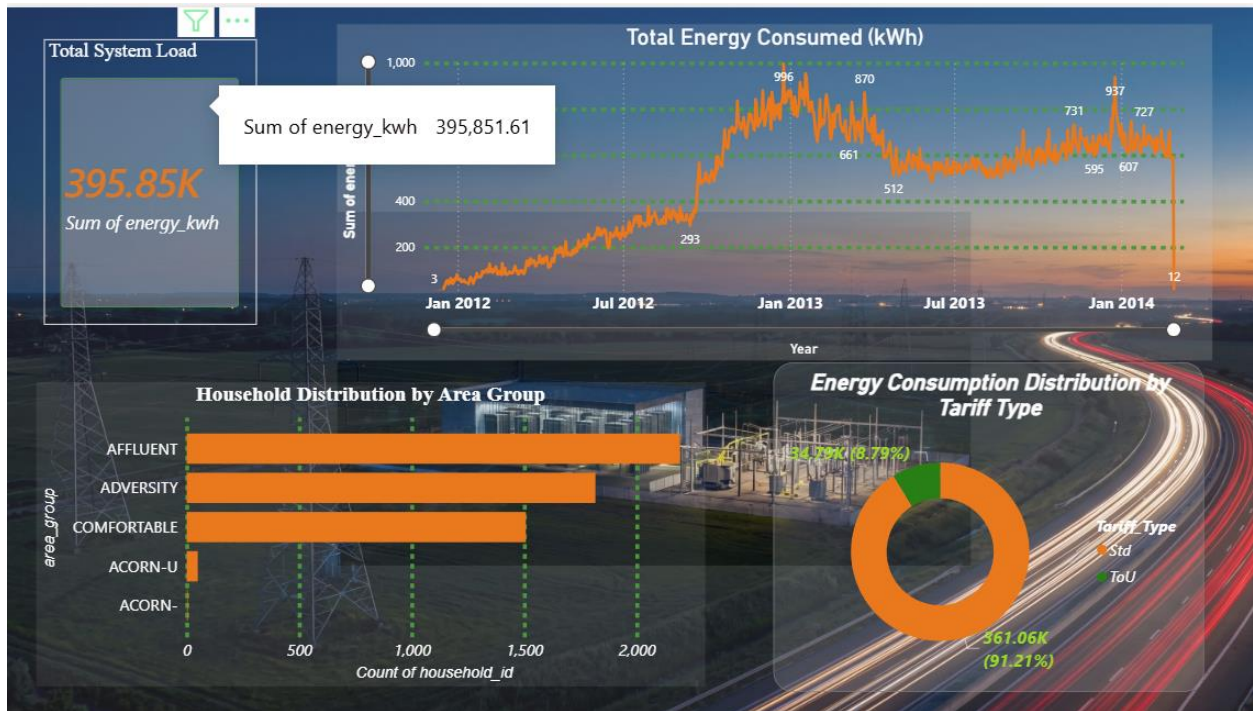
After Applying Drill Up To Visual 2



Showings Tariif Types



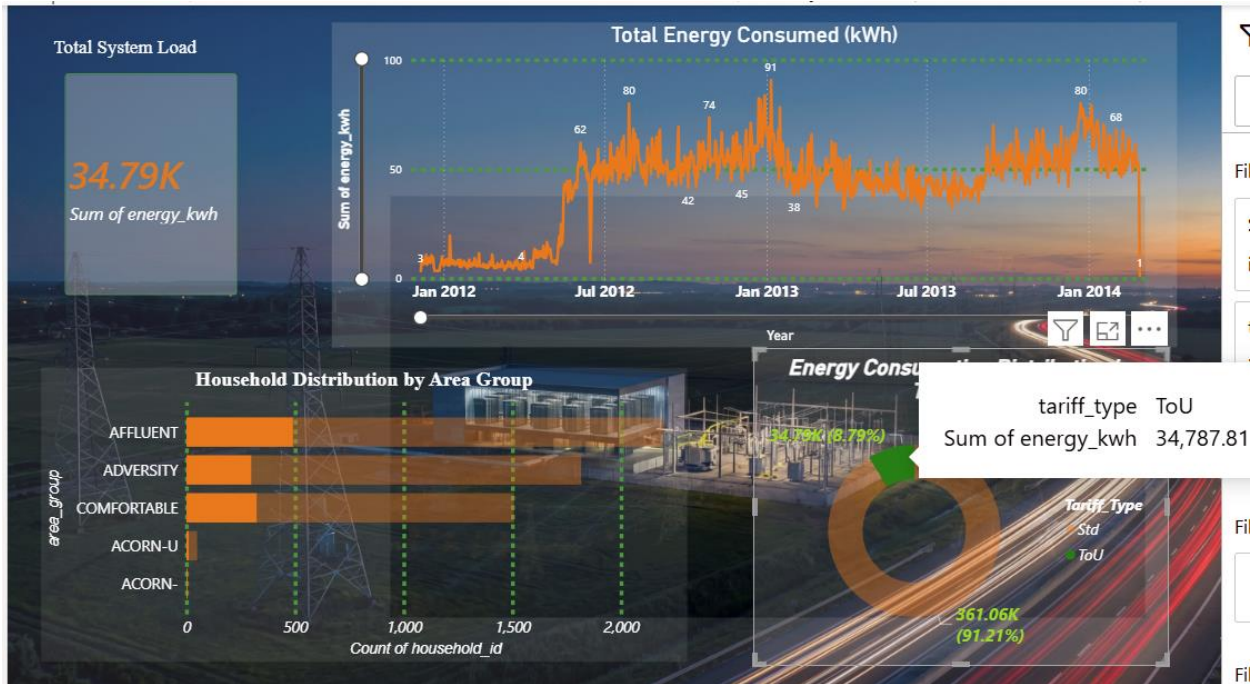
Total Demand Load On System



Total No. of Consumers By Areas Zones



ALL VISULIZATIONS BY TARIFF “ToU”

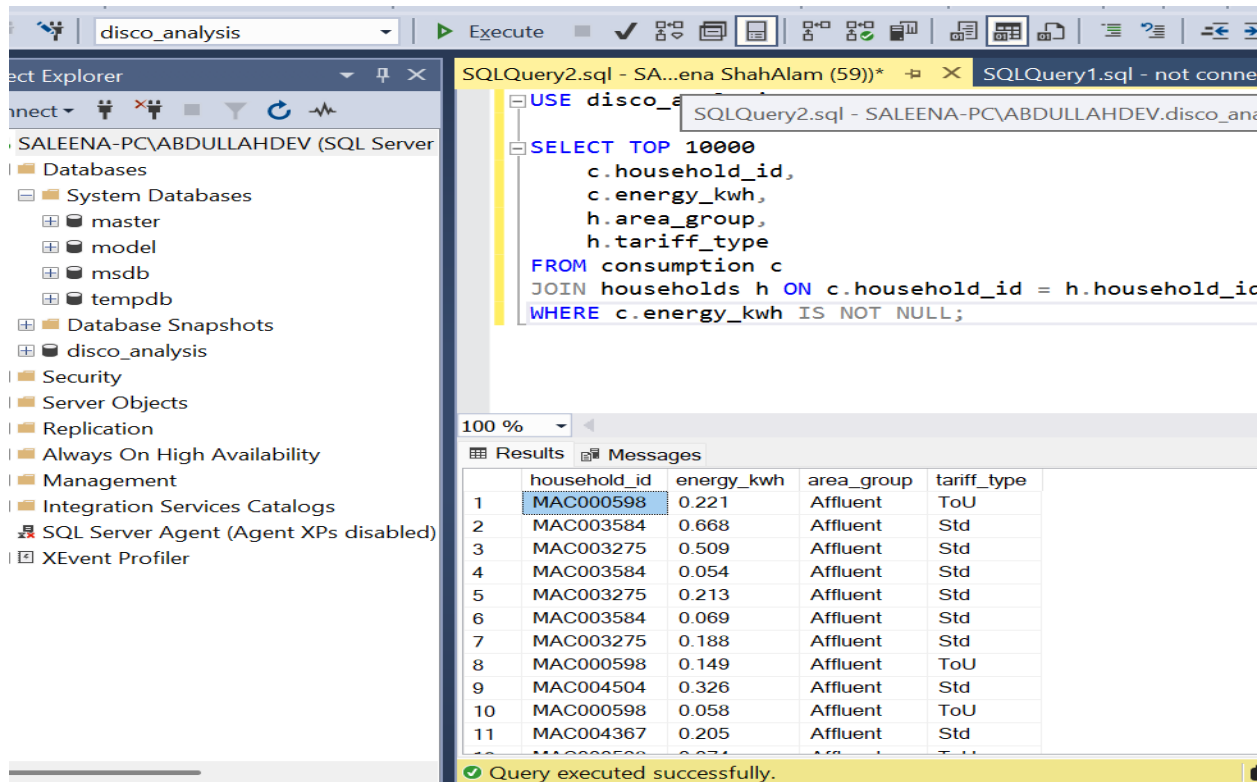


Total Power Consumption By Areas



3. ANALYSIS OF DATA (Weka)

To Generate a CSV file from Records using joins Then save results form output into CSV format.



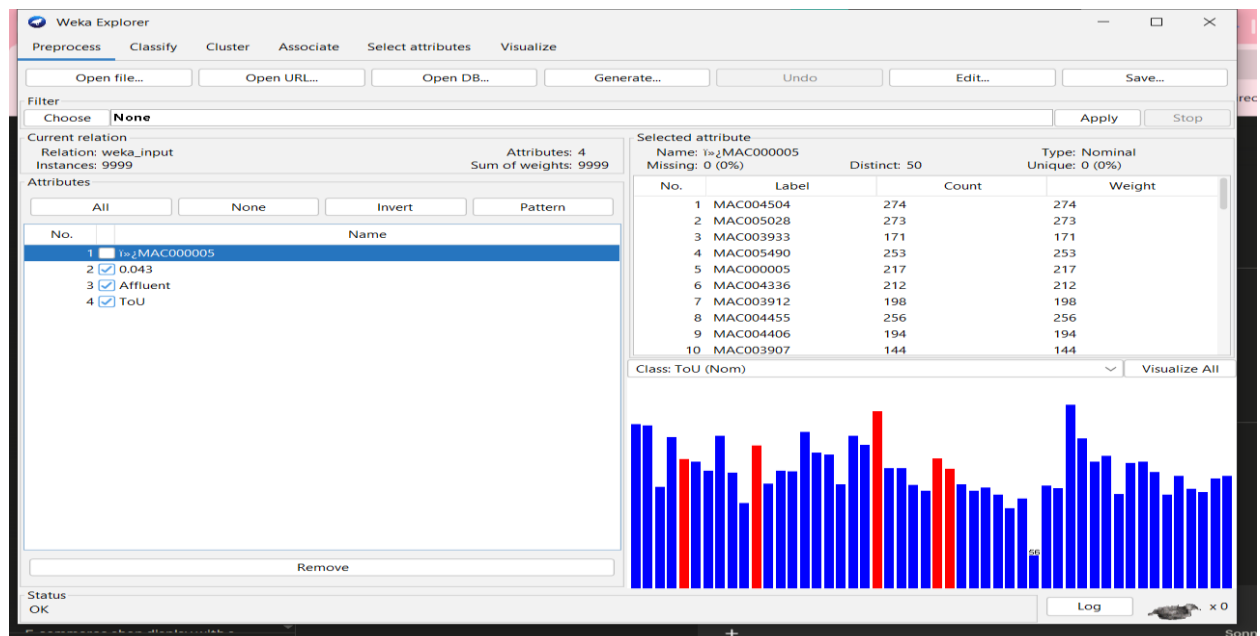
The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Object Explorer' pane displays the database structure for 'SALEENA-PC\ABDULLAHDEV (SQL Server)'. The 'disco_analysis' database is selected. The main pane shows a SQL query in 'SQLQuery2.sql':

```
USE disco_analysis
SELECT TOP 10000
    c.household_id,
    c.energy_kwh,
    h.area_group,
    h.tariff_type
FROM consumption c
JOIN households h ON c.household_id = h.household_id
WHERE c.energy_kwh IS NOT NULL;
```

Below the query, the 'Results' pane displays a table with 11 rows and 5 columns:

	household_id	energy_kwh	area_group	tariff_type
1	MAC000598	0.221	Affluent	ToU
2	MAC003584	0.668	Affluent	Std
3	MAC003275	0.509	Affluent	Std
4	MAC003584	0.054	Affluent	Std
5	MAC003275	0.213	Affluent	Std
6	MAC003584	0.069	Affluent	Std
7	MAC003275	0.188	Affluent	Std
8	MAC000598	0.149	Affluent	ToU
9	MAC004504	0.326	Affluent	Std
10	MAC000598	0.058	Affluent	ToU
11	MAC004367	0.205	Affluent	Std

A status bar at the bottom indicates 'Query executed successfully.'



The screenshot shows the Weka Explorer interface. The 'Preprocess' tab is active. The 'Current relation' is 'weka_input' with 9999 instances and 4 attributes. The 'Attributes' list shows four attributes: '0.043', 'Affluent', and 'ToU' (all checked), and '0.043' (unchecked). The 'Selected attribute' pane shows the '0.043' attribute with 50 distinct values and a nominal type. Below this, a bar chart displays the distribution of the '0.043' attribute. The chart has 50 bars, with the first 10 bars colored blue and the remaining 40 bars colored red. The y-axis represents the count of instances for each attribute value.

Simple K-Means Clustering

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Cluster

Choose SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance" -R first-last -I 500 -num-slots 1 -S 10

Cluster mode

☒ Use training set

☐ Supplied test set

☐ Percentage split % 66

☐ Classes to clusters evaluation (Nom) ToU

☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

1007:53 - SimpleKMeans

Cluster output

```
=== Run information ===
Scheme: weka.clusterers.SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance"
Relation: weka_input
Instances: 9999
Attributes: 4
1>MAC000598
0.221
Affluent
ToU
Test mode: evaluate on training data

=== Clustering model (full training set) ===

kMeans
=====
Number of iterations: 10
Within cluster sum of squared errors: 10603.964538286296

Initial starting points (random):
Cluster 0: MAC003278,0.486,Affluent,Std
Cluster 1: MAC003933,0.095,Affluent,Std

Missing values globally replaced with mean/mode

Final cluster centroids:
Attribute Full Data Cluster#
(9999.0) (1647.0) (8352.0)
=====
1>MAC000598 MAC000017 MAC004504 MAC000017
0.221 0.2801 0.8623 0.1652
Affluent Affluent Affluent Affluent
ToU Std Std Std
```

Status OK

Log x0

Algorithm Tree J48

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier

Choose J48 -C 0.25 -M 2

Test options

☐ Use training set

☐ Supplied test set

☒ Cross-validation Folds 10

☐ Percentage split % 66

More options...

(Nom) ToU

Start Stop

Result list (right-click for options)

10:10:44 - trees.J48

10:12:04 - trees.J48

Classifier output

```
1>MAC000598 = MAC004989: Std (263.0)
1>MAC000598 = MAC004712: Std (172.0)
1>MAC000598 = MAC005506: Std (201.0)
1>MAC000598 = MAC003288: Std (162.0)

Number of Leaves : 50

Size of the tree : 51

Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===
=== Summary ===

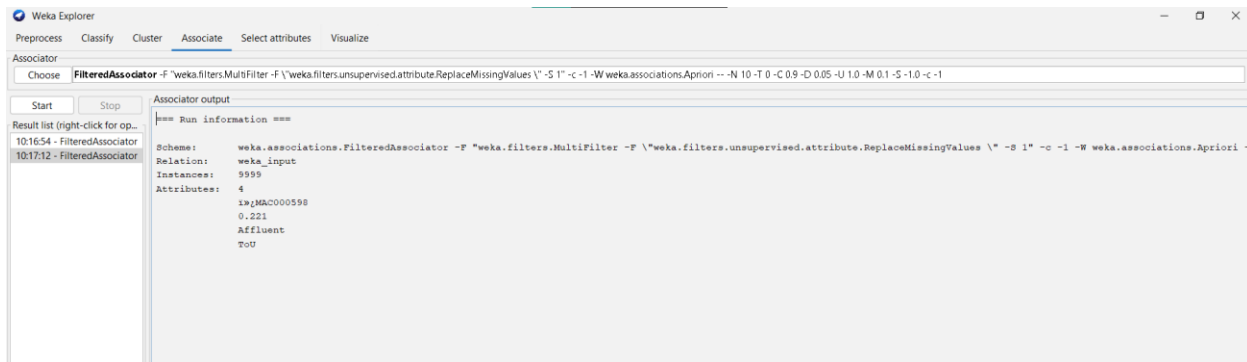
Correctly Classified Instances 9999 100 %
Incorrectly Classified Instances 0 0 %
Kappa statistic 1
Mean absolute error 0
Root mean squared error 0
Relative absolute error 0 %
Root relative squared error 0 %
Total Number of Instances 9999

=== Detailed Accuracy By Class ===

TP Rate FP Rate Precision Recall F-Measure MCC ROC Area PRC Area Class
1.000 0.000 1.000 1.000 1.000 1.000 1.000 1.000 Std
1.000 0.000 1.000 1.000 1.000 1.000 1.000 1.000 ToU
Weighted Avg. 1.000 0.000 1.000 1.000 1.000 1.000 1.000 1.000

=== Confusion Matrix ===

a b <-- classified as
8836 0 | a = Std
0 1163 | b = ToU
```



Conclusion:-

*This project successfully implemented a data-driven analysis of electricity consumption patterns using a three-tool approach. A structured data warehouse was built in MS SQL Server 2022 containing over **1.4 million smart meter readings** across household consumer zones. SQL queries revealed that Affluent zone consumers account for the highest total energy demand. Data mining using Weka's K-Means clustering segmented consumers into two distinct groups based on usage behavior, while J48 Decision Tree classification achieved 100% accuracy in predicting consumer tariff type. Power BI dashboard provided interactive visualization of monthly consumption trends, area-wise distribution, and tariff type breakdown. The findings demonstrate that data warehousing and mining techniques can effectively support DISCO load planning and consumer management decisions.*